to presage that material, briefly note the following. The value of a field at a given point in space, and the rate of change of the field's value at that point, play the same role for fields as position and velocity (momentum) play for a particle. Thus, just as we can't ever know both a definite position and a definite velocity for a particle, a field can't have a definite value and a definite rate of change of that value, at any given point in space. The more definite the field's value is at one moment, the more uncertain is the rate of change of that value—that is, the more likely it is that the field's value will change a moment later. And such change, induced by quantum uncertainty, is what I mean when referring to quantum jumps in the field's value.

11. The contribution of Linde and of Albrecht and Steinhardt was absolutely crucial, because Guth's original model—now called *old inflation*— suffered from a pernicious flaw. Remember that the supercooled Higgs field (or, in the terminology we introduce shortly, the *inflaton* field) has a value that is perched on the bump in its energy bowl *uniformly* across space. And so, while I've described how quickly the supercooled inflaton field could take the jump to the lowest energy value, we need to ask whether this quantum-induced jump would happen everywhere in space at the same time. And the answer is that it wouldn't. Instead, as Guth argued, the relaxation of the inflaton field to a zero energy value takes place by a process called bubble nucleation: the inflaton drops to its zero energy value at one point in space, and this sparks an outward-spreading bubble, one whose walls move at light speed, in which the inflaton drops to the zero energy value with the passing of the bubble wall. Guth envisioned that many such bubbles, with random centers, would ultimately coalesce to give a universe with zero-energy inflaton field everywhere. The problem, though, as Guth himself realized, was that the space surrounding the bubbles was still infused with a non-zero-energy inflaton field, and so such regions would continue to undergo rapid inflationary expansion, driving the bubbles apart. Hence, there was no guarantee that the growing bubbles would find one another and coalesce into a large, homogeneous spatial expanse. Moreover, Guth argued that the inflaton field energy was not lost as it relaxed to zero energy, but was converted to ordinary particles of matter and radiation inhabiting the universe. To achieve a model compatible with observations, though, this conversion would have to yield a *uniform*